

Sum of subsets

Saturday, 14 February 2026

8:10 PM

$$W[1:6] = \{5, 10, 12, 13, 15, 18\}$$

$$n = 6, m = 30$$

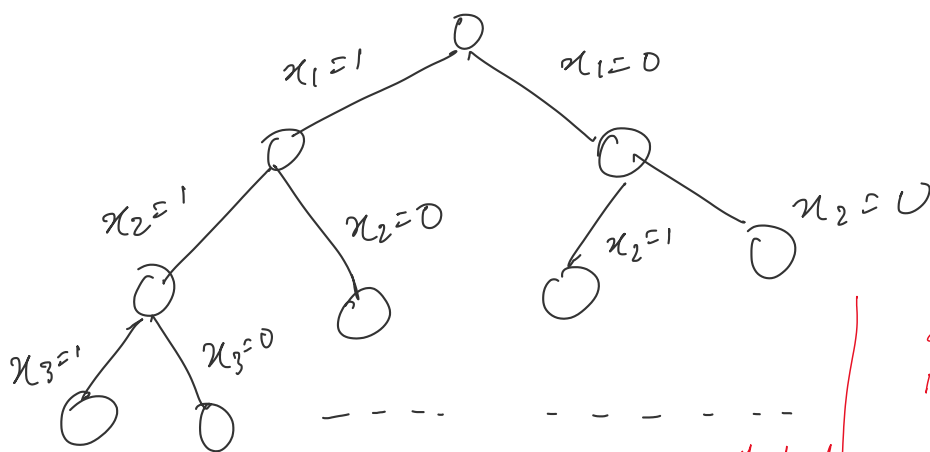
$$x = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 \end{bmatrix}$$

1 2 3 4 5 6

$x_i = 0/1$ (including or not including)

Q we have to take subsets of the weight such that their sum is exactly $= 30$

state space tree :-



total
0, 73

$x_1 = 1$

1, 20

$$\sum_{i=1}^k w_i x_i + w_{k+1} \leq m$$

$$\sum_{i=1}^k w_i x_i + \sum_{i=k+1}^n w_i > m$$

